

Associate Software Engineer

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25.00 % PERCENTAGE	FAILED Passing: 60.00%	29 / 116 SCORE	59m 59s TIME TAKEN
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TOTAL QUESTIONS	ATTEMPTED	CORRECT	WRONG	PARTIAL
30	9	3	2	4

ANSWER BREAKDOWN

#	Question	Type	Marks	Awarded	Result
1	In Go, `defer fmt.Println(i)` inside a loop captures the loop variable's CURRENT value at the moment the defer statement executes.	Single Choice	1	1	Correct
2	Two engineers disagree on inventory-system consistency: one wants SQL SERIALIZABLE isolation, the other wants application-level optimistic locking with version numbers. Argue BOTH sides with specific tradeoffs (correctness, throughput, retry semantics, deadlock vs conflict resolution). Conclude with a recommendation AND the conditions under which it would flip.	Long Answer	10	2.5	Partial
3	Design the architecture for a code-completion service backed by an LLM. Requirements: 10K concurrent users, p99 latency under 500ms, multi-region deployment, cost-efficient at scale. Cover: model serving strategy, caching layer, request batching, fallback during partial outages, observability (metrics + traces).	Long Answer	10	3	Partial
4	A long-running Python service consumes 8 GB of memory and grows ~100 MB/hour. It handles webhook callbacks and writes to a relational database. Walk through your diagnostic process: which tools you'd use, what hypotheses you'd test, and the order in which you'd apply them. List at least 5 specific tools or techniques.	Long Answer	10	3.5	Partial

#	Question	Type	Marks	Awarded	Result
5	Design a distributed rate limiter enforcing a GLOBAL limit of 1000 requests per minute across 50 API servers behind a load balancer. Cover: (a) algorithm choice (e.g., sliding window log vs token bucket vs fixed window with Redis), (b) shared state mechanism and its failure modes, (c) the consistency vs latency tradeoff during a network partition between API servers and the rate-limit store, (d) what happens to user requests when the store is unreachable.	Long Answer	10	3	Partial
6	Run-length encode a non-empty lowercase string. Group consecutive identical characters and append the count. Example: 'aaabbc' -> 'a3b2c1'. Every run gets its count appended (including runs of length 1).	Coding	8	8	Correct
7	Convert a Roman numeral to an integer. Input is a valid Roman numeral string (I, V, X, L, C, D, M). Apply subtractive notation correctly (e.g., IV=4, IX=9, XL=40, XC=90, CD=400, CM=900).	Coding	8	8	Correct
8	Given a comma-separated list of directed edges like 'A->B,B->C,C->A', return True if the graph contains a cycle, otherwise False. Empty input means no edges (False).	Coding	8	0	Wrong
9	Implement Longest Common Subsequence length. Input is a single string of the form 'A,B' where A and B are the two input strings. Return the integer length of their LCS.	Coding	8	0	Wrong